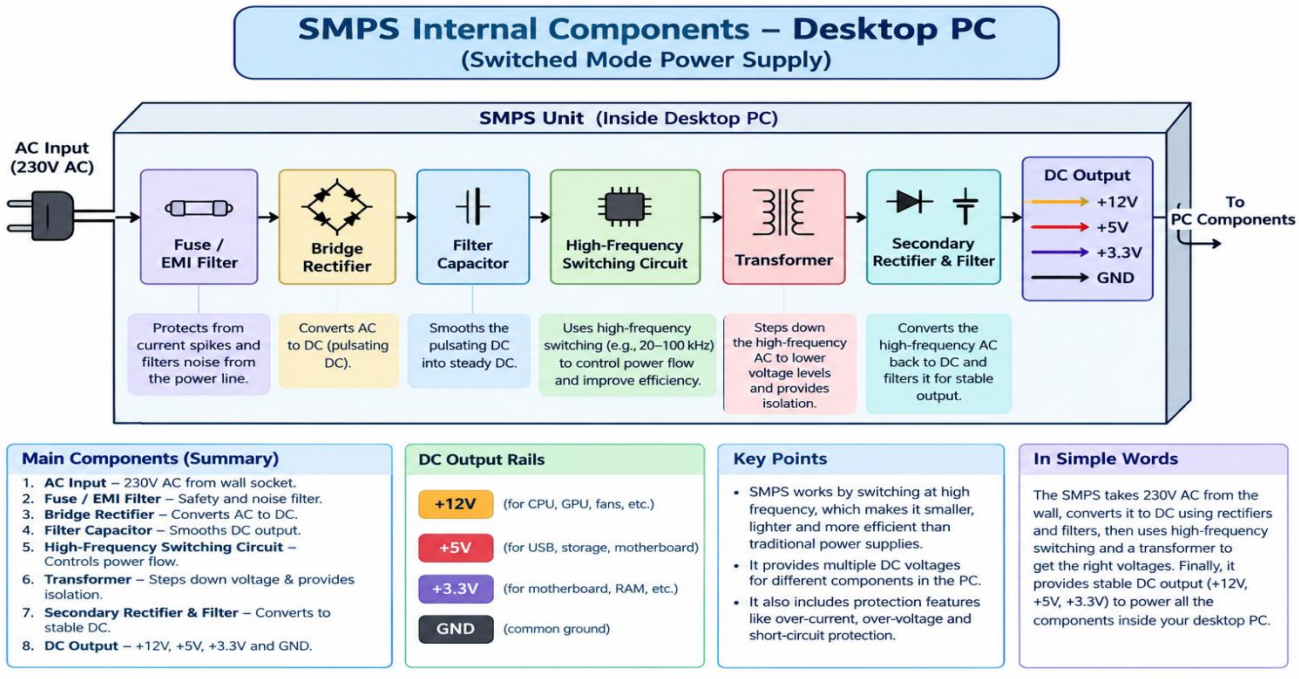
# 1: Real-World Devices Using SMPS and UPS

| Type | Device | Why it needs this power supply |
|---|---|---|
| **SMPS** | Desktop Computer | Converts AC power into the required DC voltages for PC components. |
| **SMPS** | LED/LCD Monitor | Provides stable DC power to the monitor's electronic circuits. |
| **SMPS** | Wi-Fi Router | Converts AC power into low-voltage DC power required by the router. |
| **UPS** | Desktop Computer | Provides temporary backup power during a power cut and prevents sudden shutdown. |
| **UPS** | CCTV System | Keeps cameras and recording equipment working during a power failure. |

# 2 : Diagram of a Typical Desktop PC SMPS



# 3: Power Connectors




**Name:** ATX 24-Pin Motherboard Connector

**Powers:** It supplies the main power to the **motherboard**.

**Safety Tip:** Turn off and unplug the computer before connecting or disconnecting it. Make sure the connector is correctly aligned and fully inserted.



**Name:** SATA Power Connector

**Powers:** It supplies power to SATA HDDs, SSDs, and optical drives**.**

**Safety Tip:** Align the L-shaped connector correctly and never force it into the device.



**Name:** Molex 4-Pin Connector

**Powers:** It is mainly used for older hard drives, optical drives, fans, and other older peripherals**.**

**Safety Tip:** Switch off and unplug the computer before connecting or disconnecting the Molex connector. Do not pull it by the wires.




**Name:** PCIe 6/8-Pin Power Connector

**Powers:** It provides additional power to a **dedicated graphics card (GPU)**.

**Safety Tip:** Use the **PCIe cable**, not the CPU/EPS cable, and make sure it is completely inserted into the GPU

## 4: Steps to Connect SMPS in a New Gaming PC

**Step 1: Turn off the power**

Make sure the PC is switched off and the SMPS is disconnected from the wall power.

**Step 2: Install the SMPS**

Place the SMPS inside the PC cabinet and secure it using the screws.

**Step 3: Connect the motherboard**

Connect the **24-pin ATX power cable** from the SMPS to the **24-pin motherboard power connector**.

**Step 4: Connect the CPU power**

Connect the **4+4-pin CPU/EPS cable** to the **CPU power connector** on the motherboard.

**Step 5: Connect the graphics card**

Connect the **PCIe 6-pin, 8-pin, or 6+2-pin cable**, depending on the GPU requirement, to the graphics card.

**Step 6: Connect the storage devices**

Connect a **SATA power cable** from the SMPS to the SATA SSD, HDD, or optical drive.

**Step 7: Connect other peripherals**

If required, connect **Molex power connectors** to compatible fans or older peripheral devices.

**Step 8: Check all connections**

Make sure every connector is properly inserted and locked. Do not force any connector.

**Step 9: Connect AC power**

Connect the SMPS power cable to the wall socket or a suitable UPS.

**Step 10: Turn on the PC**

Switch on the SMPS and then press the computer's power button. Check that the PC starts normally.